21.6: Comparing genome sequences provides clues to evolution and development

EVOLUTION In the last 30 years, we have seen rapid advances in genome sequencing and data collection, new techniques for assessing gene activity across the whole genome and for editing a gene sequence in a specific way in living cells, and refined approaches for understanding how genes and their products work together in complex systems. In the field of biology, we are truly poised on the brink of a new world.

Comparing genome sequences from different species reveals a lot about the evolutionary history of life, from very ancient to more recent. Similarly, comparative studies of the genetic programs that direct embryonic development in different species are uncovering the mechanisms that generated the great diversity of life-forms present today. We'll now look at what has been learned from these two approaches.

Comparing Genomes

The more similar in sequence the genes and genomes of two species are, the less time has passed for mutations and other changes to accumulate, and therefore the more closely related those species are in their evolutionary history. Comparing genomes of closely related species sheds light on more recent evolutionary events, whereas comparing genomes of very distantly related species helps us understand ancient evolutionary history. In either case, learning about characteristics that are shared or divergent between groups enhances our picture of the evolution of organisms and biological processes. Evolutionary relationships between species can be represented by a diagram in the form of a tree (often turned sideways), where each branch point marks the divergence of two lineages (see Figure 1.20). **Figure 21.17** shows the evolutionary relationships of some groups and species we'll now examine.

Figure 21.17

Evolutionary relationships of the three domains of life.

Two evolutionary tree diagrams.

The tree diagram at the top shows the ancient divergence of bacteria, archaea, and eukaryotes. A portion of the eukaryote lineage is expanded in the inset to show the more recent divergence of three mammalian species discussed in this chapter.

Parts are as follows.

Part a. Evolutionary tree diagramming relationships among 3 domains of life. A little less than 4 billion years ago is the most recent common ancestor of all living things. One branch goes to Bacteria (one domain). The second branch goes to the common ancestor (3 billion years ago) of Domains Eukarya and Archaea.

Part b. Eukaryotic lineage expanded in evolutionary tree diagramming relationships between three mammalian species (chimpanzee, human, mouse). The most recent common ancestor of the chimpanzee, human and mouse was about 65 million years ago. The lower branch goes to the mouse. The upper branch goes to the common ancestor of chimpanzees and humans (about 5 million years ago).

Comparing Distantly Related Species

Determining which genes have remained similar—that is, are *highly conserved*—in distantly related species can help clarify evolutionary relationships among species that diverged from each other long ago. Indeed, comparisons of the specific gene sequences of bacteria, archaea, and eukaryotes indicate that these three groups diverged between 2 and 4 billion years ago and strongly support the theory that they are the fundamental domains of life (see Figure 21.17).

In addition to their value in evolutionary biology, comparative genomic studies confirm the relevance of research on model organisms to our understanding of biology in general and human biology in particular. Very ancient genes can still be surprisingly similar in disparate species. One experimental study tested the ability of the human version of each of 414 important yeast genes to function equivalently in yeast cells. Remarkably, the researchers concluded that 47% of these yeast genes could be replaced by the corresponding human gene. This striking result underscores the common origin of yeasts and humans—two distantly related species.

Comparing Closely Related Species

The genomes of two closely related species are likely to be organized similarly because of their relatively recent divergence. Their long shared history also means that only a small number of gene differences are found when their genomes are compared. These genetic differences can thus be more easily correlated with phenotypic differences between the two species. An exciting application of this type of analysis is seen as researchers compare the human genome with the genomes of the chimpanzee, mouse, rat, and other mammals. Identifying the genes shared by all of these species but not by nonmammals gives us clues about what it takes to make a mammal, while finding the genes shared by chimpanzees and humans but not by rodents tells us something about primates. And, of course, comparing the human genome with that of the chimpanzee helps us answer a tantalizing question: What genomic information defines a human or a chimpanzee?

An analysis of the overall composition of the human and chimpanzee genomes, which are thought to have diverged only about 7–8 million years ago (see Figure 21.17), reveals some general differences. Considering single nucleotide substitutions, the two genomes differ by only 1.2%. When researchers looked at longer stretches of DNA, however, they were surprised to find a further 2.7% difference due to insertions or deletions of larger regions in the genome of one or the other species; many of the insertions were duplications or other repetitive DNA. In fact, a third of the human duplications are not present in the chimpanzee genome, and some of these duplications contain regions associated with human diseases. More *Alu* elements have undergone transposition in the human genome, leading to 7,000 human-specific elements compared with 2,300 chimpanzee-specific elements. The chimpanzee genome, in turn, contains many copies of a retroviral provirus not present in humans. All of these observations provide clues to the forces that might have swept the two genomes along different paths, but we don't have a complete picture yet.

Along with chimpanzees, bonobos are the other African ape species that are the closest living relatives to humans. The sequencing of the bonobo genome revealed that in some regions, human sequences were more closely related to either chimpanzee or bonobo sequences than chimpanzee or bonobo sequences were to each other. Such a fine-grained comparison of three closely related species allows even more detail to be worked out in reconstructing their related evolutionary history.

We also don't know how the genetic differences revealed by genome sequencing might account for the distinct characteristics of each species. To discover the basis for the phenotypic differences between chimpanzees and humans, biologists are studying specific genes and types of genes that differ between the two species and comparing them with their counterparts in other mammals. This approach has revealed a number of genes that are apparently changing (evolving) faster in the human than in either the chimpanzee or the mouse. Among them are genes involved in defense against malaria and tuberculosis as well as at least one gene that regulates brain size. When genes are classified by function, the genes that seem to be evolving the fastest are those that code for transcription factors. This discovery makes sense because transcription factors regulate gene expression and thus play a key role in orchestrating the overall genetic program.

One transcription factor of interest, whose gene is called FOXP2, might be involved with the acquisition of speech in humans. Attention was first drawn to FOXP2 in 2002 when a research publication reported that this gene was evolving faster in humans (*Homo sapiens*) than in other primates, based in part on two amino acids found only in the human protein sequence. The researchers proposed that this was a human-specific version of the gene that resulted from strong natural selection and might have conferred on humans the ability to speak.

In 2014, a high-quality sequence of the Neanderthal (*Homo neanderthalensis*) genome was achieved from a small amount of preserved genomic DNA, and shortly thereafter the DNA of another member of the species

Homo, the Denisovan, was sequenced as well (see Concept 34.7). It turns out that some DNA sequences from both species encode the same two amino acids, so the two amino acids are not human-specific. In a paper published in 2018, researchers analyzed genomic sequences from a larger and more diverse population of modern humans. In this more robust study, the researchers found no evidence for strong natural selection for those two amino acids in the human lineage during the time frame relevant to language acquisition.

However, the FOXP2 gene is still interesting in its own right. Several lines of evidence suggest that the FOXP2 gene product regulates genes that function in vocalization in vertebrates. First, mutations in this gene can produce severe speech and language impairment in humans. Second, the FOXP2 gene is expressed in the brains of zebra finches and canaries at the time when these songbirds are learning their songs. And third, perhaps the strongest evidence comes from a “knockout” experiment in which researchers disrupted the FOXP2 gene in mice and analyzed the resulting phenotype **(Figure 21.18)**. The homozygous mutant mice had malformed brains and failed to emit normal ultrasonic vocalizations; mice with one faulty copy of the gene also showed significant problems with vocalization. These results support the idea that the FOXP2 gene product turns on genes involved in vocalization.

Figure 21.18

Inquiry: What is the function of a gene FOXP2 that may be involved in language acquisition?

Experiment

Several lines of evidence support a role for the FOXP2 gene in the development of speech and language in humans and of vocalization in other vertebrates. In 2005, Joseph Buxbaum and collaborators at the Mount Sinai School of Medicine and several other institutions tested the function of FOXP2. They used the mouse, a model organism in which genes can be easily knocked out, as a representative vertebrate that vocalizes: Mice produce ultrasonic squeaks (whistles) to communicate stress. The researchers used genetic engineering to produce mice in which one or both copies of FOXP2 were disrupted.

Phenotypes of three mice. Wild type: two normal copies of FOXP2. Heterozygote: one copy of FOXP2 disrupted. Homozygote: both copies of FOXP2 disrupted.

They then compared the phenotypes of these mice. Two of the characters they examined are included here: brain anatomy and vocalization.

Experiment 1

Researchers cut thin sections of brain and stained them with reagents that allow visualization of brain anatomy in a UV fluorescence microscope.

Experiment 2

To induce stress, researchers separated each newborn mouse pup from its mother and recorded the number of ultrasonic whistles produced by the pup.

A photograph shows two mice. The mother mouse has a dark brown fur coat, and the newborn mouse pup has a black fur coat.

Results

Experiment 1 Results: Disruption of both copies of FOXP2 led to brain abnormalities in which the cells were disorganized. Phenotypic effects on the brain of heterozygotes, with one disrupted copy, were less severe. (Each color in the micrographs below reveals a different cell or tissue type.)

Three U V fluorescence micrographs illustrating phenotypic effects on the brain of wild type, heterozygote, and homozygote. The colors represent different cell or tissue types.

Experiment 2 Results: Disruption of both copies of FOXP2 led to an absence of ultrasonic vocalization in response to stress. The effect on vocalization in the heterozygote was also extreme.

A bar graph illustrates the results of experiment 2 of FOX P 2 genes.

The horizontal axis has the following three markings: wild type, heterozygote, and homozygote. The vertical axis labeled number of whistles ranging from 0 to 400 in increments of 100 units. The data is as follows. Wild type: 300. Heterozygote: 85. Homozygote: (No whistles).

Conclusion

FOXP2 plays a significant role in the development of functional communication systems in mice. The results augment evidence from studies of birds and humans, supporting the hypothesis that FOXP2 may act similarly in diverse organisms.

Data from W. Shu et al., Altered ultrasonic vocalization in mice with a disruption in the FOXP2 gene, *Proceedings of the National Academy of Sciences USA* 102:9643–9648 (2005).

WHAT IF? *Since the results support a role for mouse*FOXP2*in vocalization, you might wonder whether the human*FOXP2*protein is a key regulator of speech. If you were given the amino acid sequences of wild-*

*type and mutant human*FOXP2*proteins and the wild-type chimpanzee*FOXP2*protein, how would you investigate this question? What further clues could you obtain by comparing these sequences to that of the mouse*FOXP2*protein?*

*For suggested answer, see*Appendix A*.*

Expanding on this analysis, another research group replaced the FOXP2 gene in mice with a "humanized" copy coding for the human versions of the FOXP2 protein, with the two amino acids that differ between human and chimpanzee sequences. The researchers reported that although the mice with the human FOXP2 gene were generally healthy, they had subtly different vocalizations and showed changes in brain cells in circuits associated with speech in human brains. While the FOXP2 gene clearly affects vocalization and speech, the precise mechanism by which it does this is still to be determined.

The FOXP2 story is an excellent example of how different approaches can complement each other in uncovering biological phenomena of widespread importance. The FOXP2 experiments used mice as a model for humans because it would be unethical (as well as impractical) to carry out such experiments in humans. Mice and humans, which diverged about 65.5 million years ago (see Figure 21.17), share about 85% of their genes. This genetic similarity can be exploited in studying human genetic disorders. If researchers know the organ or tissue that is affected by a particular genetic disorder, they can look for genes that are expressed in these locations in mice.

Even though more distantly related to humans, fruit flies have also been a useful model species for study of such human disorders as Parkinson's disease and alcoholism, while nematodes (soil worms) have yielded a wealth of information about aging. Further research efforts are under way to extend genomic studies to many more species, including neglected species from diverse branches of the tree of life. These studies will advance our understanding of evolution, of course, as well as all aspects of biology, from human health to ecology.

Comparing Genomes Within a Species

Another exciting consequence of our ability to analyze genomes is our growing understanding of the spectrum of genetic variation in humans. Because the history of the human species is so short—probably about 200,000 years—the amount of DNA variation among humans is small compared to that of many other species. Much of our diversity seems to be in the form of single nucleotide polymorphisms (SNPs). SNPs are single base-pair sites where genetic variation is found in at least 1% of the population (see Concept 20.2); they are usually detected by DNA sequencing. In the human genome, SNPs occur on average about once in 100–300 base pairs. Scientists have already identified the locations of many tens of millions of SNP sites in the human genome and continue to find additional SNPs, especially when sequencing the whole genomes of individuals from cultural or linguistic groups not previously sampled. These are stored in databases around

the world, one of which is run by the National Center for Biotechnology Information (NCBI) and can be accessed at http://www.ncbi.nlm.nih.gov/SNP/.

In the course of this search, they have also found other variations—including chromosomal regions with inversions, deletions, and duplications. The most surprising discovery has been the widespread occurrence of *copy-number variants* (*CNVs*), loci where some individuals have one or multiple copies of a particular gene or genetic region rather than the standard two copies (one on each homolog). CNVs result from regions of the genome being duplicated or deleted inconsistently within the population. One 2020 study of 929 individuals found more than 40,000 CNVs in the genome, and these CNVs probably represent just a small subset of the total among humans. Since these variants encompass much longer stretches of DNA than the single nucleotides of SNPs, CNVs are more likely to have phenotypic consequences and to play a role in complex diseases and disorders. At the very least, the high incidence of copy-number variation blurs the meaning of the phrase "a normal human genome."

Copy-number variants, SNPs, and variations in repetitive DNA such as short tandem repeats (STRs) are useful genetic markers for studying human evolution. Several large studies over the past few years have thoroughly sequenced hundreds of individuals from diverse populations around the world—from Europe, Africa, the Middle East, Central and South Asia, East Asia, Oceania, and the Americas. Their results strongly supported earlier studies showing that genetic diversity among African genomes is far higher than among individuals from any other continent, consistent with our understanding that humans and their ancestors arose in Africa (see Concept 34.7). Further genome sequencing will continue to clarify our understanding of the migratory routes and evolution of different human populations.

Interview

Interview with Charles Rotimi: Using genomics to study health-related conditions in African-Americans

Charles Rotimi.

Widespread Conservation of Developmental Genes Among Animals

Biologists in the field of evolutionary developmental biology, or **evo-devo**, as it is often called, compare developmental processes of different multicellular organisms. Their aim is to understand how these processes have evolved and how changes in them can modify existing organismal features or lead to new ones. With the advent of molecular techniques and the recent flood of genomic information, we are beginning to realize that the genomes of related species with strikingly different forms may have only minor differences in gene sequence or, perhaps more importantly, in gene regulation. Discovering the molecular

basis of these differences in turn helps us understand the origins of the myriad diverse forms that cohabit this planet, thus informing our study of the evolution of life.

In Concept 18.4, you learned about the homeotic genes in *Drosophila melanogaster*, which encode transcription factors that regulate gene expression and specify the identity of body segments in the fruit fly (see Figure 18.20). Molecular analysis of the homeotic genes in *Drosophila* has shown that they all include a 180-nucleotide sequence called a **homeobox**, which codes for a 60-amino-acid *homeodomain* in the encoded proteins. An identical or very similar nucleotide sequence has been discovered in the homeotic genes of many invertebrates and vertebrates. The resemblance even extends to the organization of these genes: The vertebrate genes homologous to the homeotic genes of fruit flies have kept the same chromosomal arrangement **(Figure 21.19)**. The similarities in sequence and organization are so striking that one researcher has whimsically referred to flies as “little people with wings.” Homeobox-containing sequences have also been found in regulatory genes of much more distantly related eukaryotes, including plants and yeasts. From these similarities, we can deduce that the homeobox DNA sequence evolved very early and was sufficiently valuable to organisms to have been conserved in animals and plants virtually unchanged for hundreds of millions of years.

Figure 21.19

Conservation of homeotic genes in a fruit fly and a mouse.

Conservation of homeotic genes in a fruit fly and a mouse.

Homeotic genes that control the form of anterior and posterior structures of the body occur in the same linear sequence on chromosomes in Drosophila and mice. Each colored band on the chromosomes shown here represents a homeotic gene. In fruit flies, all homeotic genes are found on one chromosome. The mouse and other mammals have the same or similar sets of genes on four chromosomes. The color code indicates the parts of the embryos in which these genes are expressed and the adult body regions that result. All of these genes are essentially identical in flies and mice, except for those represented by black bands, which are less similar in the two animals.

Homeotic genes in fruit fly. Colored bands represent homeotic genes and correspond to same color body regions.

Fruit fly chromosome (described left to right, only genes described and non-coding regions omitted): Purple, black, green, black, brown, black, black, orange.

Fruit fly embryo (10 hours) (segments described left to right, anterior to posterior): purple, green, brown, (white segments), orange.

Adult fruit fly; head is green with small purple region; thorax, wings and appendages are brown; posterior end of abdomen is orange.

Homeotic genes in mouse. Colored bands represent homeotic genes and correspond to same color body regions.

Mouse chromosomes: 4 shown with corresponding colors in the same sequence lining up across chromosomes. Chromosomes described left to right, only genes described and non-coding regions omitted: First chromosome: purple, black, black, green, black, brown, black, orange, orange, orange, orange. Second chromosome: purple, black, black, green, black, brown, black, black, orange. Third chromosome: green (lines up with green on other 3 chromosomes), black, brown, black, orange, orange, orange, orange, orange. Fourth chromosome: purple, black, green, black, orange, orange, orange, orange, orange.

Mouse embryo (12 days): colored segments clustered in dorsal region. Anterior to posterior, dorsal to ventral: small purple region, longer stretch of green, brown and orange, vertebral segments green, (white), brown, (white) and orange.

Adult mouse: purple corresponds to region at base of skull; green to neck, purple to upper back, and orange to back.

Homeotic genes in animals were named *Hox* genes, short for homeobox-containing genes, because homeotic genes were the first genes found to have this sequence. Other homeobox-containing genes were later found that do not act as homeotic genes; that is, they do not directly control the identity of body parts. However, most of these genes, in animals at least, are associated with development, suggesting their ancient and fundamental importance in that process. In *Drosophila*, for example, homeoboxes are present not only in the homeotic genes but also in the egg-polarity gene *bicoid* (see Figures 18.21 and 18.22), in several of the segmentation genes, and in a master regulatory gene for eye development.

Researchers have discovered that the homeobox-encoded homeodomain is the part of the protein that binds to DNA when the protein functions as a transcription factor. Elsewhere in the protein, domains that are more variable interact with other transcription factors, allowing the homeodomain-containing protein to recognize specific enhancers and regulate the associated genes. Proteins with homeodomains probably regulate development by coordinating the transcription of batteries of developmental genes, switching them on or off. In embryos of *Drosophila* and other animal species, different combinations of homeobox genes are active in different parts of the embryo. This selective expression of regulatory genes, varying over time and space, is central to pattern formation.

Developmental biologists have found that in addition to homeotic genes, many other genes involved in development are highly conserved from species to species. These include numerous genes encoding components of signaling pathways. The extraordinary similarity among some developmental genes in

different animal species raises a question: How can the same genes be involved in the development of animals whose forms are so very different from each other?

In some cases, small changes in regulatory sequences of particular genes cause changes in gene expression patterns that can lead to major changes in body form. For example, the differing patterns of expression of the *Hox* genes along the body axis in a crustacean and an insect can explain the variation in number of leg-bearing segments among these closely related animals **(Figure 21.20)**. In other cases, similar genes direct different developmental processes in various organisms, resulting in diverse body shapes. Several *Hox* genes, for instance, are expressed in the embryonic and larval stages of the sea urchin, a nonsegmented animal that has a body plan quite different from those of insects and mice. Sea urchin adults make the pincushion-shaped shells you may have seen on the beach; two species of live sea urchins are shown in the photo. Sea urchins are among the organisms long used in classical embryological studies (see Concept 47.2).

Figure 21.20

Effect of differences in *Hox* gene expression in a crustacean and an insect.

Hox gene expression.

Changes in the expression patterns of Hox genes have occurred over evolutionary time since insects diverged from a crustacean ancestor. These changes account in part for the different body plans of ***(a)*** *the brine shrimp Artemia, a crustacean, and* ***(b)*** *the grasshopper, an insect. Shown here are regions of the adult body color-coded for expression of four Hox genes that determine the formation of particular body parts during embryonic development. Each color represents a specific Hox gene.*

Parts are as follows.

Part a. Expression of four Hox genes in the brine shrimp Artemia. Three of the Hox genes are expressed together in one region (indicated by stripes), specifying the identity of the segments that have swimmerets. The fourth (teal) specifies the identity of the genital segments.

Part b. Expression of the grasshopper version of the same four Hox genes from part a). Colored segments are thorax (purple), appendages (yellow) and abdomen (orange). Head and wings are white.

Cluster of dark colored and light purple sea urchins on the sea floor.

Two species of sea urchins

In this final chapter of the genetics unit, you have learned how studying genomic composition and comparing the genomes of different species can illuminate the process by which genomes evolve. Furthermore, comparing developmental programs, we can see that the unity of life is reflected in the similarity of molecular and cellular mechanisms used to establish body pattern, although the genes directing

development may differ among organisms. The similarities between genomes reflect the common ancestry of life on Earth. But the differences are also crucial, for they have created the huge diversity of organisms that have evolved. In the remaining chapters, we expand our perspective beyond the level of molecules, cells, and genes to explore this diversity on the organismal level.